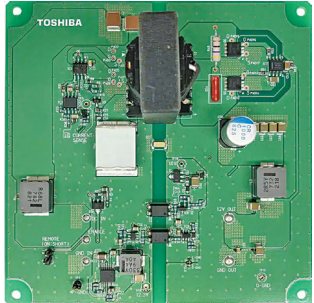


Interesting Reference Designs Of LLC RESONANT DC-DC CONVERTER

Title of Reference Design	DC/DC LLC Resonant Converter	100W LLC DC-DC Converter
Image		
Key Attractions	The 200W DC/DC converter design offers high efficiency, low switching losses, and reliable isolation, featuring full digital control, reduced component needs, and comprehensive monitoring for voltage, current, and temperature.	The 100W DC-DC converter offers an efficient 12V output with an LLC resonant circuit, achieving 88% efficiency, compact design, and enhanced performance through optimised MOSFETs and precise voltage regulation.
Highlights	The 200W DC/DC LLC resonant converter reference design by Microchip simplifies the design process for power conversion. Operating over an input range of 350V DC-420V DC, with a nominal 400V input, this design delivers a 12V DC output while ensuring voltage isolation between the primary and secondary sides. Efficiency is achieved via zero voltage switching (ZVS) on the half-bridge converter and zero current switching (ZCS) on the synchronous rectifier, further enhanced by using a synchronous rectifier instead of a full-wave rectifier. The design allows for switching techniques, such as LLC, achieving up to 95% efficiency and reducing switching losses. Features include low power consumption at no load, programmable soft-start, and monitoring and protection for voltage, current, and temperature. The design incorporates a half-bridge topology, using a resonant tank circuit with a capacitor, an inductor, and an isolating transformer. The transformer's magnetising inductance eliminates the need for a second inductor, minimising converter size, cost, and complexity, with a synchronous rectifier on the secondary side for efficiency.	The 100W LLC DC-DC converter reference design by Toshiba produces a 12V output using an LLC resonant circuit that switches the high-side and low-side MOSFETs on the primary side at a 50% duty cycle, adjusting frequency to regulate output voltage. The power supply achieves 88% efficiency at full load and has a 110mm x 110mm x 35mm footprint, excluding input and output terminals. MOSFETs and a photocoupler enhance performance and reliability, while the input voltage is stepped down by the LM46000 converter to supply 12V to the LLC controller. Input voltage range is set by external resistors dividing the input voltage to feed the enable pin of the DC-DC converter, which begins switching when this voltage exceeds 2.0V. The gate drive circuit design impacts efficiency and EMI, requiring a balance between the two. Output voltage is determined by external resistors and a shunt regulator, which adjusts the photocoupler current to match the reference voltage based on the divided output voltage.
Applications	The reference design is suitable for applications needing very low switching losses.	The reference design is suitable for industrial equipment connected to 24V output AC-DC converters and other applications.
OEM Brand	Microchip	Toshiba
URL	https://www.electronicsforu.com/electronics-projects/dc-dc-llc-resonant-converter-reference-design	https://www.electronicsforu.com/electronics-projects/100-w-llc-dc-dc-converter-reference-design